

Fourier Analytic Proofs for Wagner's k -Tree Algorithm over $\mathbb{Z}_m$

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1 Introduction

We consider the k -Sum problem over the finite abelian group $\mathbb{Z}_m$. Let $k = 2^L$, and let $L_1^{(0)}, \dots, L_k^{(0)}$ be k finite multisets of elements of $\mathbb{Z}_m$, each of cardinality approximately n . Equivalently, each list is represented by its empirical distribution, with multiplicities counted. Throughout this section, the word “list” means such a finite multiset unless explicitly stated otherwise. The objective is to find a k -tuple $(x_1, \dots, x_k)$, with each $x_j \in L_j^{(0)}$ as a multiset element, that satisfies

$$\sum_{j=1}^k x_j \equiv 0 \pmod{m}.$$

The algorithm presented, which we term the Modular k -Tree Algorithm, solves this problem using a recursive, meet-in-the-middle strategy. It operates in $L = \log_2 k$ levels. At each level i , it combines pairs of lists from the previous level $i - 1$, selects pairwise sums s satisfying $s \equiv 0 \pmod{R_i}$, and rescales these survivors into the smaller modulus $m_i = m_{i-1}/R_i$. This iterative filtering and rescaling process produces a final multiset of candidates satisfying a structured congruence modulo $D_L = \prod_{i=1}^L R_i$; the final verification step checks the reconstructed tuples against the original modulus m . The complete procedure is detailed in Algorithm 1.

Here $\text{SampleNoReplace}(M, n)$ denotes a uniformly random sample of n entries from the survivor multiset M without replacement. If the same residue appears multiple times in M with different parent pointers, these occurrences are treated as distinct multiset entries.

The correctness, or *soundness*, of Algorithm 1 is guaranteed by an invariant maintained throughout the L levels of computation. We formally state and prove this here.

Theorem 1.1 (Algorithm Soundness). *Any k -tuple $(x_1, \dots, x_k)$ returned by Algorithm 1 is a valid solution to the k -Sum problem, satisfying:*

$$\sum_{j=1}^k x_j \equiv 0 \pmod{m}$$

Proof. We prove by induction that the algorithm maintains a key invariant. Let $D_i := \prod_{t=1}^i R_t$ be the cumulative divisor up to level i . For any node v in the computation tree at level i , let $\text{leaves}(v)$ be the set of original level-0 indices for the lists that are descendants of v . For a given $y \in L_v^{(i)}$, let $(x_\ell)_{\ell \in \text{leaves}(v)}$ be the original input elements that trace to y . We define the **block sum** as $S_v := \sum_{\ell \in \text{leaves}(v)} x_\ell \pmod{m}$.

The invariant $\mathcal{I}(i)$ is: For every node v at level i and every $y \in L_v^{(i)}$, the corresponding block sum S_v satisfies:

$$S_v \equiv D_i y \pmod{m}$$

Base Case ($i = 1$): A level-1 node v is formed from a pair of level-0 lists $L_{2j-1}^{(0)}$ and $L_{2j}^{(0)}$, so $\text{leaves}(v) = \{2j-1, 2j\}$. Let the selected inputs be $a \in L_{2j-1}^{(0)}$ and $b \in L_{2j}^{(0)}$. The block sum is $S_v = (a + b) \pmod{m}$. The algorithm computes $s = (a + b) \pmod{m_0}$. Since $m_0 = m$, we have

Algorithm 1 Modular k -Tree Algorithm

Require: Modulus $m \in \mathbb{N}$, number of lists $k = 2^L$, target list size n .

Require: Level-0 finite multisets $L_1^{(0)}, \dots, L_k^{(0)}$ over $\mathbb{Z}_m$, each $|L_j^{(0)}| \approx n$.

// — Phase 1: Parameter Schedule —

1: $R_\star \leftarrow \text{round}(m^{1/(L+1)})$

2: $m_0 \leftarrow m$

3: **for** $i \leftarrow 1 \rightarrow L$ **do**

4: $R_i \leftarrow \text{divisor of } m_{i-1} \text{ closest to } R_\star$

5: $m_i \leftarrow m_{i-1}/R_i$

6: **end for**

// — Phase 2: Per-Level Filtering —

7: **for** $i \leftarrow 1 \rightarrow L$ **do**

8: $k_i \leftarrow k/2^i$

▷ Number of lists to create at this level

9: **for** $j \leftarrow 1 \rightarrow k_i$ **do**

10: $A \leftarrow L_{2j-1}^{(i-1)}$; $B \leftarrow L_{2j}^{(i-1)}$

▷ Get parent lists

11: $M \leftarrow \emptyset$

▷ Initialize multiset of survivors

12: **for all** $a \in A, b \in B$ **do**

13: $s \leftarrow (a + b) \pmod{m_{i-1}}$

14: **if** $s \equiv 0 \pmod{R_i}$ **then**

15: $y \leftarrow s/R_i$

16: Add y to M (storing parent pointers to (a, b))

17: **end if**

18: **end for**

19: **if** $|M| > n$ **then**

20: $L_j^{(i)} \leftarrow \text{SampleNoReplace}(M, n)$

21: **else**

22: $L_j^{(i)} \leftarrow M$

▷ Keep all survivors if below target

23: **end if**

24: **end for**

25: **end for**

// — Phase 3: Final Verification —

▷ Final list is $L_1^{(L)} \subseteq \mathbb{Z}_{m_L}$

26: **for** $y \in L_1^{(L)}$ **do**

27: Reconstruct $(x_1, \dots, x_k)$ from y using stored parent pointers.

28: **if** $(x_1 + \dots + x_k) \pmod{m} = 0$ **then**

29: **return** $(x_1, \dots, x_k)$

30: **end if**

31: **end for**

32: **return** FAIL

$s \equiv S_v \pmod{m}$. The algorithm filters and rescales s to $y \in L_v^{(1)}$ such that $s \equiv R_1 y \pmod{m_0}$. Since $D_1 = R_1$, we can chain the congruences:

$$S_v \equiv s \equiv R_1 y \equiv D_1 y \pmod{m}$$

Thus, $\mathcal{I}(1)$ holds.

Inductive Step ($i - 1 \rightarrow i$): Assume $\mathcal{I}(i - 1)$ holds. Let v be a level- i node formed from parent nodes u, w at level $i - 1$. Any $y \in L_v^{(i)}$ is formed from some $y_u \in L_u^{(i-1)}$ and $y_w \in L_w^{(i-1)}$. The block sum for v is $S_v = (S_u + S_w) \pmod{m}$.

By the inductive hypothesis $\mathcal{I}(i - 1)$:

$$S_u \equiv D_{i-1} y_u \pmod{m} \quad \text{and} \quad S_w \equiv D_{i-1} y_w \pmod{m}$$

Adding these gives $S_v \equiv D_{i-1}(y_u + y_w) \pmod{m}$.

From the parameter schedule, we have the identity $m = D_{i-1} m_{i-1}$. This implies that $D_{i-1} \cdot m_{i-1} \equiv 0 \pmod{m}$. Therefore, the value of $D_{i-1}(y_u + y_w) \pmod{m}$ depends only on $(y_u + y_w) \pmod{m_{i-1}}$.

The algorithm computes exactly this value: $s = (y_u + y_w) \pmod{m_{i-1}}$. Thus, we can substitute s :

$$S_v \equiv D_{i-1}(y_u + y_w) \equiv D_{i-1} s \pmod{m}$$

Next, the algorithm finds y by filtering for $s \equiv 0 \pmod{R_i}$ and rescaling, such that $s \equiv R_i y \pmod{m_{i-1}}$. Since this congruence holds modulo m_{i-1} , we can multiply by D_{i-1} and use the identity $m = D_{i-1} m_{i-1}$ to ensure the congruence holds modulo m :

$$D_{i-1} s \equiv D_{i-1}(R_i y) \pmod{m}$$

By definition, $D_i = D_{i-1} R_i$. Substituting this, we get $D_{i-1} s \equiv D_i y \pmod{m}$.

Chaining the congruences, we have $S_v \equiv D_{i-1} s \equiv D_i y \pmod{m}$. Thus, $\mathcal{I}(i)$ holds.

Conclusion: By induction, the invariant $\mathcal{I}(L)$ holds for the final list $L_1^{(L)}$. For any $y \in L_1^{(L)}$, the corresponding full k -sum $S_{\text{root}} = \sum_{j=1}^k x_j$ satisfies:

$$S_{\text{root}} \equiv D_L y \pmod{m}$$

The algorithm returns a k -tuple only if it is a valid solution. If $D_L = m$ (as in one optional setup), the invariant itself guarantees $S_{\text{root}} \equiv m y \equiv 0 \pmod{m}$. If $D_L < m$, the algorithm proceeds to the “Final Verification” step (line 28 in Algorithm 1) and only returns the tuple $(x_1, \dots, x_k)$ if it explicitly verifies that $\sum x_j \pmod{m} = 0$. In both cases, any returned tuple is a correct solution. $\square$

2 Analysis: Probability of Success

2.1 Preliminaries

We begin with a standard concentration inequality for the empirical average of complex-valued random variables, obtained by applying Hoeffding’s inequality to the real and imaginary parts [1].

Lemma 2.1 (Complex Hoeffding Bound). *Suppose $X_1, \dots, X_n$ are independent complex-valued random variables such that $|X_i|$ is always bounded by 1. If $X = \frac{1}{n} \sum_{i=1}^n X_i$ is the empirical average, then for any $\varepsilon > 0$:*

$$\mathbb{P}[|X - \mathbb{E}[X]| \geq \varepsilon] \leq 4 \exp\left(-\frac{\varepsilon^2 n}{4}\right).$$

Proof. Let $X = U + iV$, where U and V are its real and imaginary parts. The event $|X - \mathbb{E}[X]| \geq \varepsilon$ implies that either $|U - \mathbb{E}[U]| \geq \frac{\varepsilon}{\sqrt{2}}$ or $|V - \mathbb{E}[V]| \geq \frac{\varepsilon}{\sqrt{2}}$. The result follows from applying Hoeffding's inequality to U and V separately and using a union bound.

$$\begin{aligned} \mathbb{P}[|X - \mathbb{E}[X]| \geq \varepsilon] &\leq \mathbb{P}\left[|U - \mathbb{E}[U]| \geq \frac{\varepsilon}{\sqrt{2}}\right] + \mathbb{P}\left[|V - \mathbb{E}[V]| \geq \frac{\varepsilon}{\sqrt{2}}\right] \\ &\leq 4 \exp\left(-\frac{\varepsilon^2 n}{4}\right). \end{aligned}$$

□

2.2 Fourier Coefficient and Bias

Definition 2.2 (Fourier Coefficient). For a distribution f on $\mathbb{Z}_m$ and a frequency $\omega \in \mathbb{Z}_m$, the **Fourier coefficient** is defined as the expectation:

$$\hat{f}(\omega) = \mathbb{E}_{x \sim f} \left[e^{2\pi i \omega x / m} \right].$$

Fact 1 (Fourier coefficients of the uniform distribution). Let U be the uniform distribution on $\mathbb{Z}_m$, i.e., $U(x) = 1/m$ for all $x \in \mathbb{Z}_m$. Then

$$\hat{U}(\omega) = \mathbb{E}_{x \sim U} \left[e^{2\pi i \omega x / m} \right] = \begin{cases} 1, & \omega = 0, \\ 0, & \omega \neq 0. \end{cases}$$

In particular, a perfectly uniform distribution has $\hat{f}(\omega) = 0$ for all $\omega \neq 0$.

Proof. By definition,

$$\hat{U}(\omega) = \frac{1}{m} \sum_{x=0}^{m-1} e^{2\pi i \omega x / m}.$$

If $\omega = 0$, each term equals 1, so the sum is m and $\hat{U}(0) = 1$. If $\omega \neq 0$, set $r := e^{2\pi i \omega / m} \neq 1$; then the sum is a finite geometric series:

$$\sum_{x=0}^{m-1} r^x = \frac{1 - r^m}{1 - r} = \frac{1 - e^{2\pi i \omega}}{1 - r} = 0,$$

hence $\hat{U}(\omega) = 0$. □

Definition 2.3 (Bias). We measure the non-uniformity, or **bias**, of a distribution f by the largest magnitude of its non-zero Fourier coefficients:

$$\beta(f) = \max_{\omega \in \mathbb{Z}_m \setminus \{0\}} |\hat{f}(\omega)|.$$

2.3 Recombination

Lemma 2.4 (Recombination Contracts Bias). Let f_A and f_B be distributions on $\mathbb{Z}_M$ corresponding to independent draws from lists A and B . Let f_Z be the distribution of their sum $Z = A + B \pmod{M}$. Then their Fourier coefficients (Definition 2.2) multiply:

$$\widehat{f_Z}(\omega) = \widehat{f_A}(\omega) \cdot \widehat{f_B}(\omega)$$

Consequently, the bias (Definition 2.3) contracts quadratically:

$$\beta(f_Z) \leq \beta(f_A) \beta(f_B)$$

Proof. The first identity follows directly from the definition of the Fourier coefficient and the independence of A and B .

$$\begin{aligned}
\widehat{f_Z}(\omega) &= \mathbb{E}_{Z \sim f_Z} \left[e^{2\pi i \omega Z / M} \right] \\
&= \mathbb{E}_{A, B} \left[e^{2\pi i \omega (A+B) / M} \right] \\
&= \mathbb{E}_{A, B} \left[e^{2\pi i \omega A / M} \cdot e^{2\pi i \omega B / M} \right] \\
&= \mathbb{E}_{A \sim f_A} \left[e^{2\pi i \omega A / M} \right] \cdot \mathbb{E}_{B \sim f_B} \left[e^{2\pi i \omega B / M} \right] \\
&= \widehat{f_A}(\omega) \cdot \widehat{f_B}(\omega)
\end{aligned}$$

The second inequality follows by taking the magnitude of the coefficients. For any $\omega \neq 0$, by Definition 2.3:

$$|\widehat{f_Z}(\omega)| = |\widehat{f_A}(\omega)| \cdot |\widehat{f_B}(\omega)| \leq \beta(f_A) \cdot \beta(f_B)$$

This holds for all non-zero frequencies, so it must also hold for the maximum, which completes the proof. $\square$

2.4 Subgroup Indicator

The algorithm's selection step at level i (operating modulo $M = m_{i-1}$) filters elements based on their membership in the subgroup $H_i \subseteq \mathbb{Z}_M$. This subgroup is defined as the set of all elements divisible by the level's divisor, R_i :

$$H_i = \{x \in \mathbb{Z}_M : x \equiv 0 \pmod{R_i}\}$$

This selection can be modeled as a multiplication of the distribution by the indicator function $\mathbf{1}_{H_i}(x)$, defined as:

$$\mathbf{1}_{H_i}(x) = \begin{cases} 1 & \text{if } x \in H_i \\ 0 & \text{if } x \notin H_i \end{cases}$$

The **annihilator** of H_i , denoted $H_i^\perp$, is the subgroup of frequencies that evaluate to 1 on all elements of H_i :

$$H_i^\perp = \left\{ \eta \in \mathbb{Z}_M : e^{2\pi i \eta h / M} = 1 \forall h \in H_i \right\}$$

Given $H_i = \{x \in \mathbb{Z}_M : x \equiv 0 \pmod{R_i}\}$, the annihilator subgroup is precisely:

$$H_i^\perp = \left\{ 0, \frac{M}{R_i}, 2\frac{M}{R_i}, \dots, (R_i - 1)\frac{M}{R_i} \right\}$$

Claim 2.5 (Support of Subgroup Indicator). *The unnormalized Fourier transform of the indicator function $\mathbf{1}_{H_i}$, defined as*

$$\widehat{\mathbf{1}_{H_i}}(\omega) = \sum_{x \in \mathbb{Z}_M} \mathbf{1}_{H_i}(x) e^{2\pi i \omega x / M}$$

is supported only on the annihilator $H_i^\perp$.

Proof. We can rewrite the transform as a sum only over the subgroup H_i :

$$\widehat{\mathbf{1}_{H_i}}(\omega) = \sum_{h \in H_i} e^{2\pi i \omega h / M}$$

We consider two cases.

Case 1: $\omega \in H_i^\perp$. By the definition of the annihilator, $e^{2\pi i \omega h/M} = 1$ for all $h \in H_i$. The sum becomes:

$$\widehat{\mathbf{1}_{H_i}}(\omega) = \sum_{h \in H_i} 1 = |H_i| = \frac{M}{R_i}$$

This is non-zero.

Case 2: $\omega \notin H_i^\perp$. By definition, there must exist some $h_0 \in H_i$ such that $e^{2\pi i \omega h_0/M} \neq 1$. Since H_i is a subgroup, summing over $h \in H_i$ is equivalent to summing over $h' = h + h_0$, as h' also traverses the entire subgroup.

$$\begin{aligned} \widehat{\mathbf{1}_{H_i}}(\omega) &= \sum_{h \in H_i} e^{2\pi i \omega h/M} = \sum_{h \in H_i} e^{2\pi i \omega (h+h_0)/M} \\ &= \sum_{h \in H_i} \left(e^{2\pi i \omega h/M} \cdot e^{2\pi i \omega h_0/M} \right) \\ &= e^{2\pi i \omega h_0/M} \sum_{h \in H_i} e^{2\pi i \omega h/M} \\ \widehat{\mathbf{1}_{H_i}}(\omega) &= e^{2\pi i \omega h_0/M} \cdot \widehat{\mathbf{1}_{H_i}}(\omega) \end{aligned}$$

Rearranging the terms, we get:

$$\widehat{\mathbf{1}_{H_i}}(\omega) \cdot \left(1 - e^{2\pi i \omega h_0/M} \right) = 0$$

Since we are in Case 2, the term in the parenthesis is non-zero. Therefore, it must be that $\widehat{\mathbf{1}_{H_i}}(\omega) = 0$.

Thus, the transform is non-zero if and only if $\omega \in H_i^\perp$. $\square$

Our goal is to understand how the Fourier coefficients (Definition 2.2) of a distribution f are affected by this subgroup filtering, i.e., what is the transform of the new (unnormalized) distribution $f \cdot \mathbf{1}_{H_i}$. The following lemma about the “alias-average” identity provides the exact relationship.

Lemma 2.6 (Fourier Transform of Filtered Distribution). *Let f be a distribution on $\mathbb{Z}_M$, and let H_i be the subgroup $\{x \in \mathbb{Z}_M : x \equiv 0 \pmod{R_i}\}$. The Fourier transform of the filtered function $f \cdot \mathbf{1}_{H_i}$ is given by:*

$$\widehat{f \cdot \mathbf{1}_{H_i}}(\omega) = \frac{1}{R_i} \sum_{t=0}^{R_i-1} \hat{f}\left(\omega + t \frac{M}{R_i}\right)$$

Proof. The proof relies on expressing the indicator function $\mathbf{1}_{H_i}$ as an average of characters from its annihilator subgroup, $H_i^\perp$. We previously established (Claim 2.5) that this subgroup is $H_i^\perp = \{t \cdot (M/R_i) : t = 0, \dots, R_i - 1\}$. The key identity is:

$$\mathbf{1}_{H_i}(x) = \frac{1}{R_i} \sum_{t=0}^{R_i-1} e^{2\pi i (tM/R_i)x/M} = \frac{1}{R_i} \sum_{t=0}^{R_i-1} e^{2\pi i t x / R_i}$$

This holds because if $x \in H_i$, then x is a multiple of R_i , so $e^{2\pi i t x / R_i} = 1$ for all t . The sum is R_i , and the average is 1. If $x \notin H_i$, this is a geometric series of a non-trivial root of unity, which sums to 0.

We now compute the Fourier transform of the product by substituting this identity:

$$\begin{aligned}
\widehat{f \cdot \mathbf{1}_{H_i}}(\omega) &= \sum_{x \in \mathbb{Z}_M} f(x) \cdot \mathbf{1}_{H_i}(x) \cdot e^{2\pi i \omega x / M} \\
&= \sum_{x \in \mathbb{Z}_M} f(x) \left(\frac{1}{R_i} \sum_{t=0}^{R_i-1} e^{2\pi i t x / R_i} \right) e^{2\pi i \omega x / M} \\
&= \frac{1}{R_i} \sum_{t=0}^{R_i-1} \sum_{x \in \mathbb{Z}_M} f(x) \cdot e^{2\pi i (tM/R_i)x / M} \cdot e^{2\pi i \omega x / M} \\
&= \frac{1}{R_i} \sum_{t=0}^{R_i-1} \sum_{x \in \mathbb{Z}_M} f(x) \cdot e^{2\pi i (\omega + tM/R_i)x / M}
\end{aligned}$$

We recognize the inner sum as the definition of the Fourier coefficient (Definition 2.2), $\hat{f}$, evaluated at the frequency $\left(\omega + t\frac{M}{R_i}\right)$:

$$\widehat{f \cdot \mathbf{1}_{H_i}}(\omega) = \frac{1}{R_i} \sum_{t=0}^{R_i-1} \hat{f}\left(\omega + t\frac{M}{R_i}\right).$$

□

2.5 Distribution after Renormalization and Rescaling

The “alias-average” identity from Lemma 2.6 describes the transform of the unnormalized list of survivors. The algorithm, however, performs two subsequent steps:

1. **Renormalization:** The filtered distribution $f \cdot \mathbf{1}_{H_i}$ is not a probability distribution (it sums to $P \leq 1$). We must re-normalize it by dividing by the total probability of selection, $P = \Pr_{x \sim f}[x \in H_i]$. Let this new, valid distribution on $\mathbb{Z}_M$ be $f'(x) = (f(x) \cdot \mathbf{1}_{H_i}(x))/P$.
2. **Rescaling:** The algorithm maps survivors $x \in H_i$ to a new, smaller group $\mathbb{Z}_{m_i}$ (where $m_i = M/R_i$) via the isomorphism $y = x/R_i$.

Let g be the final distribution on $\mathbb{Z}_{m_i}$. For any $y \in \mathbb{Z}_{m_i}$, its probability $g(y)$ is the probability of its antecedent $x = yR_i$ in the renormalized distribution f' . Thus, $g(y) = f'(yR_i) = f(yR_i)/P$.

The following lemma establishes the crucial link between the Fourier transform of this final, rescaled distribution g and the original distribution f .

Lemma 2.7 (Transform after Rescaling). *Let f be a distribution on $\mathbb{Z}_M$, and let g be the new distribution on $\mathbb{Z}_{m_i}$ (where $m_i = M/R_i$) obtained by conditioning on the subgroup $H_i = \{x : x \equiv 0 \pmod{R_i}\}$ and rescaling by $y = x/R_i$. Let $P = \Pr_{x \sim f}[x \in H_i]$ be the probability of selection, and assume $P > 0$.*

For any frequency $\omega' \in \mathbb{Z}_{m_i}$, the Fourier coefficient $\hat{g}(\omega')$ is given by:

$$\hat{g}(\omega') = \frac{1}{P \cdot R_i} \sum_{t=0}^{R_i-1} \hat{f}(\omega' + t \cdot m_i)$$

Proof. We first compute the Fourier transform of g over its domain $\mathbb{Z}_{m_i}$, using the relationship

$$g(y) = f(yR_i)/P:$$

$$\begin{aligned}\hat{g}(\omega') &= \mathbb{E}_{y \sim g} \left[e^{2\pi i \omega' y / m_i} \right] = \sum_{y=0}^{m_i-1} g(y) \cdot e^{2\pi i \omega' y / m_i} \\ &= \sum_{y=0}^{m_i-1} \left(\frac{f(yR_i)}{P} \right) \cdot e^{2\pi i \omega' y / m_i} \\ &= \frac{1}{P} \sum_{y=0}^{m_i-1} f(yR_i) \cdot e^{2\pi i \omega' y / m_i}\end{aligned}$$

Now, let $f'(x) = (f(x) \cdot \mathbf{1}_{H_i}(x))/P$ be the renormalized distribution on the original group $\mathbb{Z}_M$. Its transform $\hat{f}'(\omega)$ is supported only on H_i , so we can sum over $x = yR_i$:

$$\begin{aligned}\hat{f}'(\omega) &= \sum_{x \in \mathbb{Z}_M} f'(x) \cdot e^{2\pi i \omega x / M} = \sum_{y=0}^{m_i-1} f'(yR_i) \cdot e^{2\pi i \omega (yR_i) / M} \\ &= \sum_{y=0}^{m_i-1} \left(\frac{f(yR_i)}{P} \right) \cdot e^{2\pi i \omega y (R_i / M)}\end{aligned}$$

Using the identity $R_i/M = 1/m_i$, this becomes:

$$\hat{f}'(\omega) = \frac{1}{P} \sum_{y=0}^{m_i-1} f(yR_i) \cdot e^{2\pi i \omega y / m_i}$$

By comparing the expressions for $\hat{g}(\omega')$ and $\hat{f}'(\omega)$, we see $\hat{g}(\omega') = \hat{f}'(\omega')$ for any $\omega' \in \mathbb{Z}_{m_i}$.

From the linearity of the Fourier transform and Lemma 2.6, we have:

$$\hat{f}'(\omega') = \frac{1}{P} \widehat{f \cdot \mathbf{1}_{H_i}}(\omega') = \frac{1}{P} \left[\frac{1}{R_i} \sum_{t=0}^{R_i-1} \hat{f} \left(\omega' + t \frac{M}{R_i} \right) \right]$$

Substituting $\hat{g}(\omega') = \hat{f}'(\omega')$ and $m_i = M/R_i$ gives the final result. $\square$

Before we can analyze the bias of the new distribution g , we must first understand the selection probability P , as it appears in the denominator of the transform in Lemma 2.7. The following claim shows that P is also controlled by the Fourier coefficients of f .

Claim 2.8 (Selection Probability). *The probability of an element $x \sim f$ falling into the subgroup H_i , denoted $P = \Pr_{x \sim f}[x \in H_i]$, is determined by the Fourier coefficients of f on the annihilator subgroup $H_i^\perp$:*

$$P = \frac{1}{R_i} \sum_{t=0}^{R_i-1} \hat{f}(t \cdot m_i)$$

where $m_i = M/R_i$.

Proof. By definition, $P = \sum_{x \in H_i} f(x) = \sum_{x \in \mathbb{Z}_M} f(x) \cdot \mathbf{1}_{H_i}(x)$. This is precisely the $\omega = 0$ Fourier coefficient of the filtered function $f \cdot \mathbf{1}_{H_i}$. Using the “alias-average” identity from Lemma 2.6 and setting $\omega = 0$:

$$P = \widehat{f \cdot \mathbf{1}_{H_i}}(0) = \frac{1}{R_i} \sum_{t=0}^{R_i-1} \hat{f} \left(0 + t \frac{M}{R_i} \right) = \frac{1}{R_i} \sum_{t=0}^{R_i-1} \hat{f}(t \cdot m_i) \quad \square$$

This claim shows that if the bias $\beta(f)$ (Definition 2.3) is small, P must be very close to the uniform probability $1/R_i$. This is the final component needed to prove our main lemma for this section, which shows that the entire selection and rescaling process preserves small bias.

Lemma 2.9 (Selection and Rescaling Preserves Small Bias). *Let g be the rescaled distribution from Lemma 2.7 and f be the input distribution with bias $\beta(f) \leq \varepsilon$.*

1. *For any non-zero frequency $\omega' \in \mathbb{Z}_{m_i} \setminus \{0\}$, the new Fourier coefficients are bounded:*

$$|\hat{g}(\omega')| \leq \frac{1}{P \cdot R_i} \sum_{t=0}^{R_i-1} |\hat{f}(\omega' + t \cdot m_i)| \leq \frac{\beta(f)}{P}$$

2. *If $\varepsilon < 1/R_i$, the selection probability P is close to the uniform probability,*

$$\left| P - \frac{1}{R_i} \right| \leq \varepsilon$$

and the bias of the new distribution g is bounded:

$$\beta(g) \leq \frac{\varepsilon}{\frac{1}{R_i} - \varepsilon} = R_i \varepsilon \cdot (1 + O(R_i \varepsilon))$$

Proof. Part 1: The first inequality is a direct application of the triangle inequality ($|\sum z_i| \leq \sum |z_i|$) to the result of Lemma 2.7.

$$|\hat{g}(\omega')| = \left| \frac{1}{P \cdot R_i} \sum_{t=0}^{R_i-1} \hat{f}(\omega' + t \cdot m_i) \right| \leq \frac{1}{P \cdot R_i} \sum_{t=0}^{R_i-1} |\hat{f}(\omega' + t \cdot m_i)|$$

For the second inequality, we note that $\omega' \neq 0 \pmod{m_i}$. This implies that all frequencies $\omega_t = \omega' + t \cdot m_i$ in the sum are non-zero modulo M . By the definition of bias (Definition 2.3), $|\hat{f}(\omega_t)| \leq \beta(f)$ for all t .

$$\frac{1}{P \cdot R_i} \sum_{t=0}^{R_i-1} |\hat{f}(\omega_t)| \leq \frac{1}{P \cdot R_i} \sum_{t=0}^{R_i-1} \beta(f) = \frac{1}{P \cdot R_i} (R_i \cdot \beta(f)) = \frac{\beta(f)}{P}$$

Part 2: We use Claim 2.8 to bound P . The $t = 0$ term is $\hat{f}(0) = 1$. All other $R_i - 1$ terms correspond to non-zero frequencies, so their magnitudes are bounded by $\beta(f) \leq \varepsilon$.

$$P = \frac{1}{R_i} \left(\hat{f}(0) + \sum_{t=1}^{R_i-1} \hat{f}(t \cdot m_i) \right)$$

By the triangle inequality:

$$\left| P - \frac{1}{R_i} \right| = \left| \frac{1}{R_i} \sum_{t=1}^{R_i-1} \hat{f}(t \cdot m_i) \right| \leq \frac{1}{R_i} \sum_{t=1}^{R_i-1} |\hat{f}(t \cdot m_i)| \leq \frac{1}{R_i} (R_i - 1) \varepsilon < \varepsilon$$

This establishes the bound on P , which implies $P \geq \frac{1}{R_i} - \varepsilon$. Finally, we substitute this into the result from Part 1. For any $\omega' \neq 0$:

$$|\hat{g}(\omega')| \leq \frac{\beta(f)}{P} \leq \frac{\varepsilon}{\frac{1}{R_i} - \varepsilon}$$

Since this holds for all $\omega' \neq 0$, it also holds for the maximum, $\beta(g)$. □

2.6 Subsampling and Additive Error

The algorithm restricts the size of intermediate lists to n by taking a uniform random subset of the survivors (lines 21-22 in Algorithm 1). While the expected distribution of the random subset matches the original list, the finite sampling process injects a statistical additive error into the Fourier coefficients.

Lemma 2.10 (Subsampling Injects Additive Bias - Single Frequency). *Let M be a finite list of survivors in $\mathbb{Z}_{m_i}$ with $|M| > n$, and let f_M be its empirical distribution. Let f_S be the empirical distribution of a list S formed by taking a uniform random sample of M of size n without replacement. For any fixed frequency $\omega \in \mathbb{Z}_{m_i}$ and any $\nu > 0$:*

$$\mathbb{P}\left(|\widehat{f_S}(\omega) - \widehat{f_M}(\omega)| \geq \nu\right) \leq 4 \exp\left(-\frac{n\nu^2}{4}\right).$$

Proof. Let $X_1, \dots, X_n$ be the n random samples drawn without replacement from M . The empirical Fourier coefficient for f_S at frequency ω is the sample average:

$$\widehat{f_S}(\omega) = \frac{1}{n} \sum_{j=1}^n e^{2\pi i \omega X_j / m_i}.$$

Let $Z_j = e^{2\pi i \omega X_j / m_i}$. Because each draw has an equal marginal probability of being any element in the finite population M , the expected value of the sample mean exactly matches the population mean:

$$\mathbb{E}[\widehat{f_S}(\omega)] = \frac{1}{|M|} \sum_{x \in M} e^{2\pi i \omega x / m_i} = \widehat{f_M}(\omega).$$

The variables Z_j are complex-valued and bounded strictly by 1 (since $|e^{i\theta}| = 1$). Although the draws $X_1, \dots, X_n$ are not independent, Hoeffding's comparison theorem for sampling without replacement implies that the tail probability of the sample mean is bounded above by the corresponding with-replacement tail probability [1, Theorem 4].

Consequently, the independent-case Complex Hoeffding Bound (Lemma 2.1) serves as a valid, conservative upper bound for our sample:

$$\mathbb{P}\left(|\widehat{f_S}(\omega) - \mathbb{E}[\widehat{f_S}(\omega)]| \geq \nu\right) \leq 4 \exp\left(-\frac{n\nu^2}{4}\right),$$

which completes the proof. $\square$

Corollary 2.11 (Global Subsampling Noise Floor). *Let $\delta \in (0, 1)$ be the target failure probability for the subsampling phase. Let the additive noise floor be defined as:*

$$\nu = 2\sqrt{\frac{\ln(4km/\delta)}{n}}.$$

With probability at least $1 - \delta$, the additive error introduced by random sampling without replacement for all $k - 1$ internal lists across the entire computation tree is uniformly bounded by ν . Consequently, the bias of any subsampled list S from a parent list M satisfies $\beta(f_S) \leq \beta(f_M) + \nu$.

Proof. The computation tree creates exactly $k - 1$ internal lists that are subject to subsampling. The number of non-zero frequencies evaluated at any level i is $m_i - 1 < m$.

We apply a global union bound over all these internal lists and their non-zero frequencies. The total number of evaluated events is strictly less than $k \cdot m$. By Lemma 2.10, the probability that any single frequency deviates by more than ν is $4 \exp(-n\nu^2/4)$. Setting the union bound of all failure probabilities for the subsampling phase to be at most δ :

$$km \cdot 4 \exp\left(-\frac{n\nu^2}{4}\right) \leq \delta \implies \exp\left(-\frac{n\nu^2}{4}\right) \leq \frac{\delta}{4km}.$$

Taking the natural logarithm and solving for ν yields the stated noise floor.

When this global condition holds, for any internal list S subsampled from M and any $\omega \neq 0$, the triangle inequality yields:

$$|\widehat{f_S}(\omega)| \leq |\widehat{f_M}(\omega)| + |\widehat{f_S}(\omega) - \widehat{f_M}(\omega)| \leq \beta(f_M) + \nu.$$

Taking the maximum over all $\omega \neq 0$ confirms the bound on the bias: $\beta(f_S) \leq \beta(f_M) + \nu$. $\square$

Lemma 2.12 (Noisy Bias Recurrence and Steady State). *Let $R = \max_{j \in [L]} R_j$ be the maximum divisor across all levels in Algorithm 1. Fix a total failure probability $\delta \in (0, 1)$ and define*

$$\nu_\delta := 2\sqrt{\frac{\ln(8km/\delta)}{n}}.$$

Let ε_i denote the maximum empirical bias of any level- i list after the optional size cap has been applied. Assume $\nu_\delta \leq 1/(8R)$. Then, with probability at least $1 - \delta$, the following statements hold simultaneously for all levels $i \in \{0, \dots, L\}$:

$$\varepsilon_i \leq 2R_i \varepsilon_{i-1}^2 + \nu_\delta \leq 2R \varepsilon_{i-1}^2 + \nu_\delta \quad (i \geq 1),$$

and

$$\varepsilon_i \leq 2\nu_\delta.$$

Proof. Let $\mathcal{E}_{\text{init}}$ be the event from Corollary 2.14 with failure probability $\delta/2$, and let $\mathcal{E}_{\text{sub}}$ be the event from Corollary 2.11 with failure probability $\delta/2$. By the union bound,

$$\Pr[\mathcal{E}_{\text{init}} \cap \mathcal{E}_{\text{sub}}] \geq 1 - \delta.$$

We condition on this intersection for the rest of the proof.

On $\mathcal{E}_{\text{init}}$,

$$\varepsilon_0 \leq 2\sqrt{\frac{\ln(8k(m-1)/\delta)}{n}} < 2\sqrt{\frac{\ln(8km/\delta)}{n}} = \nu_\delta,$$

so $\varepsilon_0 \leq 2\nu_\delta$.

Now assume inductively that $\varepsilon_{i-1} \leq 2\nu_\delta$. Recombining two level- $(i-1)$ empirical distributions gives an intermediate sum distribution with bias at most ε_{i-1}^2 by Lemma 2.4. Applying Lemma 2.9 to the subsequent filtering and rescaling step gives the pre-sampling level- i bias bound

$$\frac{\varepsilon_{i-1}^2}{1/R_i - \varepsilon_{i-1}^2}.$$

This application is valid because

$$\varepsilon_{i-1} \leq 2\nu_\delta \leq \frac{1}{4R}, \quad \text{hence} \quad \varepsilon_{i-1}^2 \leq \frac{1}{16R^2} \leq \frac{1}{2R_i},$$

where the last inequality uses $R_i \leq R$ and $R \geq 1$. Therefore

$$\frac{1}{R_i} - \varepsilon_{i-1}^2 \geq \frac{1}{2R_i},$$

and consequently

$$\frac{\varepsilon_{i-1}^2}{1/R_i - \varepsilon_{i-1}^2} \leq 2R_i \varepsilon_{i-1}^2.$$

On $\mathcal{E}_{\text{sub}}$, the optional size cap at level i adds at most ν_δ to every nonzero Fourier coefficient. Thus

$$\varepsilon_i \leq 2R_i \varepsilon_{i-1}^2 + \nu_\delta \leq 2R \varepsilon_{i-1}^2 + \nu_\delta.$$

Using the induction hypothesis once more,

$$\varepsilon_i \leq 2R(2\nu_\delta)^2 + \nu_\delta = 8R\nu_\delta^2 + \nu_\delta.$$

Since $\nu_\delta \leq 1/(8R)$, we have $8R\nu_\delta^2 \leq \nu_\delta$, so $\varepsilon_i \leq 2\nu_\delta$. This closes the induction. $\square$

2.7 Initial Bias of Random Lists

We assume the initial lists $L_i^{(0)}$ are finite multisets generated by sampling n elements uniformly and independently from $\mathbb{Z}_m$. The Fourier coefficients of the corresponding empirical distribution $f_{L_i^{(0)}}$ are random variables. For any $\omega \neq 0$, the expectation is $\mathbb{E}[\hat{f}_{L_i^{(0)}}(\omega)] = 0$. The following claims establish that the initial empirical biases are small with high probability.

Claim 2.13 (Single-List, Single-Frequency Bound). *Let $f_{L^{(0)}}$ be the empirical distribution of a list $L^{(0)}$ formed by n i.i.d. uniform samples from $\mathbb{Z}_m$. For any single non-zero frequency $\omega \in \mathbb{Z}_m \setminus \{0\}$ and any $\varepsilon > 0$:*

$$\mathbb{P}\left(|\hat{f}_{L^{(0)}}(\omega)| \geq \varepsilon\right) \leq 4 \exp\left(-\frac{n\varepsilon^2}{4}\right)$$

Proof. Let $X_1, \dots, X_n$ be the i.i.d. uniform samples. The empirical coefficient is $\hat{f}_{L^{(0)}}(\omega) = \frac{1}{n} \sum_{j=1}^n e^{2\pi i \omega X_j/m}$. Let $Z_j = e^{2\pi i \omega X_j/m}$. For $\omega \neq 0$, $\mathbb{E}[Z_j] = 0$, so $\mathbb{E}[\hat{f}_{L^{(0)}}(\omega)] = 0$. Each Z_j is a complex variable with $|Z_j| = 1$. Then we can conclude the proof by invoking the Complex Hoeffding Bound (Lemma 2.1). $\square$

Corollary 2.14 (Initial Bias Bound). *Let*

$$\varepsilon_0 = \max_{i \in [k]} \beta(f_{L_i^{(0)}})$$

be the maximum empirical bias over all k initial lists. With probability at least $1 - \delta$,

$$\varepsilon_0 \leq 2 \sqrt{\frac{\ln(4k(m-1)/\delta)}{n}}.$$

Proof. This follows by applying a union bound to Claim 2.13 over all k lists and all $m-1$ non-zero frequencies.

$$\mathbb{P}(\varepsilon_0 \geq \varepsilon) \leq k \cdot (m-1) \cdot 4 \exp\left(-\frac{n\varepsilon^2}{4}\right)$$

Setting this failure probability to δ and solving for ε gives the result. $\square$

Implications for Parameters. This bound informs the analysis. With high probability, the initial bias of our lists is $\varepsilon_0 = \Theta\left(\sqrt{\frac{\log(km)}{n}}\right)$.

- After the first recombination step (summing two independent lists), the bias squares (Lemma 2.4), becoming $\beta(f_Z) \leq \varepsilon_0^2 = \Theta\left(\frac{\log(km)}{n}\right)$.
- At the first selection step (level 1), we apply Lemma 2.9. The new bias is bounded by $\beta(g) \leq R_1 \varepsilon_0^2 \cdot (1 + O(R_1 \varepsilon_0^2))$.

2.8 Final Success Probability

The preceding bounds control the empirical Fourier bias of the lists produced by Algorithm 1. We now translate a final empirical bias bound into a success probability for the actual final sampling rule of the algorithm.

Let M_L denote the final survivor multiset at level L before the optional size cap is applied, and let f_{M_L} be its empirical distribution on $\mathbb{Z}_{m_L}$. Let

$$H := |M_L|, \quad Z := |\{y \in M_L : y = 0\}|, \quad \theta := Z/H$$

when $H > 0$. Finally, let $N_L := \min\{n, H\}$ be the number of candidates examined after the final size cap.

By the invariant in Theorem 1.1, if a final candidate has rescaled value $y = 0 \in \mathbb{Z}_{m_L}$, then its reconstructed tuple has full sum congruent to $D_L \cdot 0 \equiv 0 \pmod{m}$. Thus every zero element in the final rescaled multiset is a genuine solution candidate, and the final verification step will accept it.

Bias to empirical point mass. Assume $H > 0$ and suppose

$$\beta(f_{M_L}) \leq \varepsilon_L.$$

Fourier inversion on $\mathbb{Z}_{m_L}$ gives

$$\theta = f_{M_L}(0) = \frac{1}{m_L} \sum_{\omega \in \mathbb{Z}_{m_L}} \widehat{f_{M_L}}(\omega) = \frac{1}{m_L} \left(1 + \sum_{\omega \neq 0} \widehat{f_{M_L}}(\omega) \right).$$

Therefore

$$\left| \theta - \frac{1}{m_L} \right| \leq \frac{1}{m_L} \sum_{\omega \neq 0} \left| \widehat{f_{M_L}}(\omega) \right| \leq \frac{m_L - 1}{m_L} \varepsilon_L < \varepsilon_L.$$

In particular,

$$\theta \geq \frac{1}{m_L} - \varepsilon_L.$$

Lemma 2.15 (Final sampling from an empirical survivor pool). *Condition on the final survivor multiset M_L . If $H = |M_L|$ and $\theta = Z/H$, then the final sampling-and-verification step of Algorithm 1 succeeds with conditional probability at least*

$$1 - (1 - \theta)^{N_L}, \quad N_L = \min\{n, H\}.$$

Proof. If $H \leq n$, the algorithm keeps all final survivors. If $Z = 0$, both sides are zero. If $Z \geq 1$, the algorithm sees a zero and succeeds, while $1 - (1 - \theta)^H \leq 1$.

It remains to consider $H > n$. Then $N_L = n$, and the final list is obtained by sampling n entries without replacement from M_L . The probability of missing all Z zero entries is

$$\frac{(H - Z)_{N_L}}{(H)_{N_L}} = \prod_{q=0}^{N_L-1} \frac{H - Z - q}{H - q},$$

with the convention that the numerator is zero if $N_L > H - Z$. For each factor with nonzero numerator,

$$\frac{H - Z - q}{H - q} = 1 - \frac{Z}{H - q} \leq 1 - \frac{Z}{H} = 1 - \theta.$$

Thus the miss probability is at most $(1 - \theta)^{N_L}$, proving that the hit probability is at least $1 - (1 - \theta)^{N_L}$. $\square$

Proposition 2.16 (Actual Final-List Success Bound). *Condition on a final survivor multiset M_L with $H > 0$. If its empirical distribution satisfies*

$$\beta(f_{M_L}) \leq \varepsilon_L < \frac{1}{m_L},$$

then the conditional probability that Algorithm 1 finds a solution in its final verification step is at least

$$\Pr[\text{Success} \mid M_L] \geq 1 - \left(1 - \frac{1}{m_L} + \varepsilon_L \right)^{N_L}, \quad N_L = \min\{n, H\}.$$

Proof. The bias-to-point-mass estimate gives $\theta \geq 1/m_L - \varepsilon_L > 0$. Lemma 2.15 then implies

$$\Pr[\text{Success} \mid M_L] \geq 1 - (1 - \theta)^{N_L}.$$

The function $u \mapsto 1 - (1 - u)^{N_L}$ is increasing on $[0, 1]$, so substituting the lower bound on θ yields

$$\Pr[\text{Success} \mid M_L] \geq 1 - \left(1 - \frac{1}{m_L} + \varepsilon_L\right)^{N_L}.$$

□

The balanced final-sampling regime. The bound is already constant once the number of final candidates examined is on the order of m_L . For example, if $N_L \geq m_L$ and $\varepsilon_L \leq c/m_L$ for some fixed $0 < c < 1$, then Proposition 2.16 gives

$$\Pr[\text{Success} \mid M_L] \geq 1 - \left(1 - \frac{1 - c}{m_L}\right)^{N_L} \geq 1 - \left(1 - \frac{1 - c}{m_L}\right)^{m_L}.$$

Using $(1 - x/r)^r \leq e^{-x}$ for $x > 0$, this implies

$$\Pr[\text{Success} \mid M_L] \geq 1 - e^{-(1-c)}.$$

Thus $N_L = \Omega(m_L)$ final examined candidates are sufficient for constant conditional success probability. If the target cap n is much larger than m_L , the lower bound only improves, provided the final survivor pool contains at least that many candidates.

2.9 A Sufficient List Size Exponent

We now derive a sufficient target list size n that keeps the Fourier noise small enough for Proposition 2.16. Let the level divisors satisfy $R_i \leq R$, with the intended schedule $R \approx m^{1/(L+1)}$, and write

$$n = R^c$$

for a constant exponent c .

Step 1: Steady-state and final-bias constraints. By Lemma 2.12, with total failure probability δ , the global noise floor is

$$\nu_\delta = 2\sqrt{\frac{\ln(8km/\delta)}{n}}.$$

The steady-state recurrence requires

$$\nu_\delta \leq \frac{1}{8R}.$$

On the corresponding good event, every level- $(L - 1)$ list has empirical bias at most $2\nu_\delta$.

Let M_L be the level- L survivor multiset before the optional final size cap. It is obtained by recombining two level- $(L - 1)$ lists, filtering, and rescaling. Therefore its empirical bias satisfies

$$\varepsilon_L \leq 2R_L(2\nu_\delta)^2 \leq 8R\nu_\delta^2.$$

To make Proposition 2.16 usable with, say, $\varepsilon_L \leq 1/(2m_L)$, it is sufficient that

$$8R\nu_\delta^2 \leq \frac{1}{2m_L}.$$

Equivalently,

$$\nu_\delta \leq \frac{1}{4\sqrt{Rm_L}}.$$

Thus it suffices to impose

$$\nu_\star := \min \left\{ \frac{1}{8R}, \frac{1}{4\sqrt{Rm_L}} \right\}, \quad \nu_\delta \leq \nu_\star.$$

Step 2: Solving for c . Substituting $n = R^c$ into the definition of ν_δ , the condition $\nu_\delta \leq \nu_\star$ becomes

$$2\sqrt{\frac{\ln(8km/\delta)}{R^c}} \leq \nu_\star.$$

Squaring and rearranging gives

$$R^c \geq \frac{4 \ln(8km/\delta)}{\nu_\star^2}.$$

Hence a sufficient exponent is

$$c \geq \frac{\ln(4 \ln(8km/\delta)) - 2 \ln(\nu_\star)}{\ln R}.$$

Step 3: Evaluating the two constraints. There are two cases.

- **Steady-state constraint.** If $\nu_\star = 1/(8R)$, then

$$R^c \geq 256R^2 \ln(8km/\delta),$$

so it is sufficient that

$$c \geq 2 + \frac{\ln(256 \ln(8km/\delta))}{\ln R}.$$

- **Final-bias constraint.** If $\nu_\star = 1/(4\sqrt{Rm_L})$, then

$$R^c \geq 64Rm_L \ln(8km/\delta),$$

so it is sufficient that

$$c \geq 1 + \frac{\ln m_L}{\ln R} + \frac{\ln(64 \ln(8km/\delta))}{\ln R}.$$

Under the intended parameter schedule $m_L \approx R$, both constraints have leading exponent 2, up to logarithmic finite-size corrections. Thus the analysis requires

$$n \gtrsim R^2 \text{polylog}(km/\delta)$$

to control the Fourier noise.

This is not the same as the balanced final-sampling condition $N_L = \Omega(m_L)$ from Section 2.8. The cap $n = R^c$ controls the empirical Fourier error introduced by intermediate sampling. The final success bound requires only that the number N_L of final candidates examined be on the order of m_L ; when $n \gg m_L$ and the final survivor pool is large enough, the final hit probability bound improves rather than worsens.

3 A Single-List Variant

We now analyze a simplified single-list version of the modular k -tree algorithm. In contrast to Section 2, all $k = 2^L$ leaves are drawn from one initial multiset L_0 , and the algorithm keeps every survivor at every intermediate level; there is no intermediate subsampling.

The proof in this section uses an exact prime-power parameter schedule. Fix an odd prime q , set

$$R = q, \quad m = q^{L+1}, \quad m_i = q^{L+1-i} \quad (0 \leq i \leq L).$$

Thus $m_0 = m$, $m_i = m_{i-1}/R$, and $m_L = q = R$. This exact divisibility is not merely a notational convenience: the filtering step is conditioning on the subgroup $R\mathbb{Z}_{m_{i-1}} \subseteq \mathbb{Z}_{m_{i-1}}$, and the rescaling map $x \mapsto x/R$ identifies this subgroup with $\mathbb{Z}_{m_i}$. The Fourier identities from Section 2, and the soundness invariant from Theorem 1.1, rely on this exact subgroup structure.

For an arbitrary target scale M , one may choose a prime $q \asymp M^{1/(L+1)}$ and run the exactly divisible schedule $m' = q^{L+1}$. Bertrand's postulate guarantees such a prime within a constant factor of $M^{1/(L+1)}$, so for fixed L this changes the relevant asymptotic exponents only by constant factors. This observation should be understood as a complexity-preserving exact schedule, not as a proof for an arbitrary prescribed modulus M . The theorem below is stated and proved only for the exact schedule $m = q^{L+1}$.

The initial list L_0 consists of

$$n = \lfloor \alpha q \rfloor$$

independent uniform samples from $\mathbb{Z}_m$, where $\alpha > 0$ is a fixed constant. The algorithm forms ordered pairs at each level. Therefore the same initial sample may appear in several different candidate trees, but our lower bound will count only candidate trees whose k leaves use distinct initial sample indices. Since these candidates are a subset of the candidates examined by the algorithm, this gives a valid lower bound on the actual success probability.

Algorithm 2 Modular Single-List k -Tree Algorithm

Require: Odd prime q , number of elements $k = 2^L$.

Require: Modulus $m = q^{L+1}$, divisor $R = q$, and level moduli $m_i = q^{L+1-i}$.

Require: Level-0 multiset $L_0 = (X_1, \dots, X_n)$ of size $n = \lfloor \alpha q \rfloor$, with entries in $\mathbb{Z}_m$.

Ensure: A valid zero-sum combination of $k = 2^L$ elements from L_0 .

// — Phase 1: Exact Parameter Schedule —

1: $R \leftarrow q$

2: $m_0 \leftarrow q^{L+1}$

3: **for** $i \leftarrow 1 \rightarrow L$ **do**

4: $m_i \leftarrow m_{i-1}/R$

▷ Equivalently, $m_i = q^{L+1-i}$

5: **end for**

// — Phase 2: Per-Level Filtering —

6: **for** $i \leftarrow 1 \rightarrow L$ **do**

7: $L_i \leftarrow \emptyset$

▷ Initialize multiset of survivors

8: **for all** ordered pairs of multiset entries $a, b \in L_{i-1}$ **do**

9: $s \leftarrow (a + b) \pmod{m_{i-1}}$

10: **if** $s \equiv 0 \pmod{R}$ **then**

▷ Efficiently found via meet-in-the-middle

11: $y \leftarrow s/R$

12: Add y to L_i (storing parent pointers to (a, b))

▷ All survivors are kept; no

subsampling

13: **end if**

14: **end for**

15: **end for**

// — Phase 3: Final Verification —

▷ Final list is $L_L \subseteq \mathbb{Z}_{m_L}$

16: **for** $y \in L_L$ **do**

17: Reconstruct $(x_1, \dots, x_k)$ from y using stored parent pointers.

18: **if** $(x_1 + \dots + x_k) \pmod{m} = 0$ **then**

19: **return** $(x_1, \dots, x_k)$

20: **end if**

21: **end for**

22: **return** FAIL

3.1 Candidate Trees and Zero-Admissibility

Let $\mathcal{T}_L$ be the complete binary tree with $k = 2^L$ leaves. We measure height from the leaves: leaves have height 0, their parents have height 1, and the root has height L . For a node v , let

$\text{Leaves}(v)$ denote the set of leaf positions below v .

Let the initial list be

$$L_0 = (X_1, \dots, X_n),$$

where $X_1, \dots, X_n$ are independent uniform random variables in $\mathbb{Z}_m$, with $m = q^{L+1}$. A leaf-index assignment is a vector

$$\mathbf{a} = (a_1, \dots, a_k) \in [n]^k.$$

It assigns the value X_{a_j} to the j th leaf of $\mathcal{T}_L$. For a node v , define the block sum

$$S_v(\mathbf{a}) = \sum_{j \in \text{Leaves}(v)} X_{a_j} \in \mathbb{Z}_m.$$

Definition 3.1 (Surviving and zero-admissible assignments). A leaf-index assignment $\mathbf{a} \in [n]^k$ is called surviving if, for every internal node v of height $h(v) \in \{1, \dots, L\}$,

$$S_v(\mathbf{a}) \equiv 0 \pmod{q^{h(v)}}.$$

It is called zero-admissible if it satisfies the same conditions at every proper internal node and, at the root r ,

$$S_r(\mathbf{a}) \equiv 0 \pmod{q^{L+1}}.$$

The distinction is that a surviving assignment reaches the final list $L_L \subseteq \mathbb{Z}_{m_L} = \mathbb{Z}_q$, whereas a zero-admissible assignment reaches the final list with final value $0 \in \mathbb{Z}_q$.

Whenever a residue class $S \in \mathbb{Z}_{q^{L+1}}$ satisfies $S \equiv 0 \pmod{q^i}$, we write

$$\frac{S}{q^i} \pmod{q^{L+1-i}}$$

for the unique residue class $Y \in \mathbb{Z}_{q^{L+1-i}}$ satisfying

$$S \equiv q^i Y \pmod{q^{L+1}}.$$

Thus all divisions by powers of q below are divisions inside the corresponding quotient group, not divisions of a chosen integer representative.

Lemma 3.2 (Tree constraints match Algorithm 2). *A leaf-index assignment $\mathbf{a}$ reaches level i at a node v of height i if and only if*

$$S_u(\mathbf{a}) \equiv 0 \pmod{q^{h(u)}}$$

for every internal descendant u of v with height $h(u) \leq i$. In this case, the value stored at v is

$$Y_v(\mathbf{a}) \equiv \frac{S_v(\mathbf{a})}{q^i} \pmod{q^{L+1-i}}.$$

Consequently, $\mathbf{a}$ appears in the final list with value 0 if and only if it is zero-admissible.

Proof. We prove the statement by induction on the height i .

For $i = 1$, a node v has two leaf children with values X_{a_j} and $X_{a_{j'}}$. Algorithm 2 keeps the ordered pair precisely when

$$X_{a_j} + X_{a_{j'}} \equiv 0 \pmod{q},$$

which is exactly the condition $S_v(\mathbf{a}) \equiv 0 \pmod{q}$. The stored value is

$$Y_v(\mathbf{a}) \equiv \frac{X_{a_j} + X_{a_{j'}}}{q} = \frac{S_v(\mathbf{a})}{q} \pmod{q^L}.$$

Assume the claim holds for height $i - 1$. Let v be a node of height i , with children u, w of height $i - 1$. By induction, the values stored at u, w , if they exist, satisfy

$$Y_u \equiv \frac{S_u}{q^{i-1}}, \quad Y_w \equiv \frac{S_w}{q^{i-1}} \pmod{q^{L+2-i}}.$$

Algorithm 2 keeps their ordered pair at node v precisely when

$$Y_u + Y_w \equiv 0 \pmod{q}.$$

Substituting the inductive expressions, this is equivalent to

$$\frac{S_u + S_w}{q^{i-1}} \equiv 0 \pmod{q},$$

which is equivalent to

$$S_v = S_u + S_w \equiv 0 \pmod{q^i}.$$

When this condition holds, the value stored at v is

$$Y_v \equiv \frac{Y_u + Y_w}{q} \equiv \frac{S_u + S_w}{q^i} = \frac{S_v}{q^i} \pmod{q^{L+1-i}}.$$

This completes the induction.

At the root, $i = L$ and the final stored value lies in $\mathbb{Z}_{m_L} = \mathbb{Z}_q$. The final value is 0 exactly when

$$\frac{S_r(\mathbf{a})}{q^L} \equiv 0 \pmod{q},$$

equivalently

$$S_r(\mathbf{a}) \equiv 0 \pmod{q^{L+1}}.$$

Thus final value 0 is exactly zero-admissibility. $\square$

3.2 A Conditional Probability Lemma

We next prove the only probabilistic estimate needed for the second-moment argument. The point is that every still-random leaf contributes at least one factor of q^{-1} to the probability of producing a zero at the root.

Consider a complete binary tree of height $h \leq L$. Some of its leaves may be fixed to arbitrary values of $\mathbb{Z}_m$, and the remaining r leaves are independent uniform random variables in $\mathbb{Z}_m$. Let S_h denote the event that every internal node v of height $a \leq h$ has block sum divisible by q^a . On S_h , define the root output digit

$$\Psi_h = \frac{S_{\text{root}}}{q^h} \pmod{q}.$$

Lemma 3.3 (Survival and zero probability with fixed leaves). *For every $h \leq L$ and every fixing of some leaves:*

$$\Pr[S_h] \leq q^{-\max\{r-1, 0\}}.$$

Moreover, if $r \geq 1$ and $\Pr[S_h] > 0$, then conditioned on S_h , the digit Ψ_h is uniform on $\mathbb{Z}_q$. Consequently, for every fixing,

$$\Pr[S_h \text{ and } \Psi_h = 0] \leq q^{-r},$$

where the case $r = 0$ is interpreted as the trivial bound $\Pr[S_h \wedge \{\Psi_h = 0\}] \leq 1$.

Finally, in the special case where no leaves are fixed, so that $r = 2^h$, both bounds are exact:

$$\Pr[S_h] = q^{-(r-1)} \quad \text{and} \quad \Pr[S_h \text{ and } \Psi_h = 0] = q^{-r}.$$

Proof. We first prove the conditional uniformity statement. Assume $r \geq 1$ and choose one unfixed leaf. For any $c \in \mathbb{Z}_q$, modify this leaf value by adding cq^h modulo $m = q^{L+1}$. This map is a bijection on the set of assignments to the unfixed leaves.

Consider any internal node of height $a < h$. Its block sum changes either by 0 or by cq^h , and $q^a \mid q^h$, so its divisibility condition modulo q^a is preserved. The root block sum changes by cq^h , so the root condition modulo q^h is also preserved. However,

$$\Psi_h = \frac{S_{\text{root}}}{q^h} \pmod{q}$$

is shifted by c . Hence, for every $b \in \mathbb{Z}_q$, the satisfying assignments with $\Psi_h = b$ are in bijection with the satisfying assignments with $\Psi_h = b + c$. Therefore, if $\Pr[\mathcal{S}_h] > 0$, the conditional distribution of Ψ_h given $\mathcal{S}_h$ is uniform on $\mathbb{Z}_q$.

We next prove the survival upper bound by induction on h .

For $h = 0$, there are no internal constraints, so $\Pr[\mathcal{S}_0] = 1$. Since a height-0 tree has either $r = 0$ or $r = 1$, this equals $q^{-\max\{r-1, 0\}}$.

Now let $h \geq 1$. Let the left and right subtrees contain r_1 and r_2 unfixed leaves, so $r = r_1 + r_2$. Let $\mathcal{S}_{h-1}^L$ and $\mathcal{S}_{h-1}^R$ denote the survival events inside the two subtrees. These events depend on disjoint sets of random leaves, hence are independent. By induction,

$$\Pr[\mathcal{S}_{h-1}^L] \leq q^{-\max\{r_1-1, 0\}}, \quad \Pr[\mathcal{S}_{h-1}^R] \leq q^{-\max\{r_2-1, 0\}}.$$

If $\Pr[\mathcal{S}_{h-1}^L] = 0$ or $\Pr[\mathcal{S}_{h-1}^R] = 0$, then by independence $\Pr[\mathcal{S}_h] \leq \Pr[\mathcal{S}_{h-1}^L \wedge \mathcal{S}_{h-1}^R] = 0$, and the claimed bound holds trivially. Assume from now on that both subtree survival probabilities are positive, so that the conditional-uniformity statement applies to each subtree.

If both $r_1, r_2 > 0$, then, conditional on the two subtree survival events, the two subtree output digits are independent and uniform on $\mathbb{Z}_q$ by the uniformity statement above. The root survival condition is exactly that these two digits sum to 0 modulo q , which has conditional probability $1/q$. Therefore

$$\Pr[\mathcal{S}_h] \leq q^{-(r_1-1)} q^{-(r_2-1)} q^{-1} = q^{-(r-1)}.$$

If exactly one of r_1, r_2 is positive, say $r_1 = r$ and $r_2 = 0$, then the root condition has conditional probability at most 1, so

$$\Pr[\mathcal{S}_h] \leq q^{-(r-1)}.$$

Remark. The conditional-uniformity statement applied to the nontrivial subtree shows that the root condition in fact has conditional probability exactly $1/q$, yielding the sharper $\Pr[\mathcal{S}_h] \leq q^{-r}$. We retain the weaker bound $q^{-(r-1)}$ because it suffices for the inductive hypothesis and matches the bound in Case 1.

If $r_1 = r_2 = 0$, then the event is deterministic and

$$\Pr[\mathcal{S}_h] \leq 1 = q^{-\max\{r-1, 0\}}.$$

This proves the survival upper bound.

If $r \geq 1$, the uniformity statement gives

$$\Pr[\mathcal{S}_h \wedge \{\Psi_h = 0\}] = \Pr[\mathcal{S}_h] \cdot \frac{1}{q} \leq q^{-(r-1)} q^{-1} = q^{-r}.$$

If $r = 0$, the displayed bound reads at most 1, which is trivial.

It remains to prove exactness when no leaves are fixed. We prove

$$\Pr[\mathcal{S}_h] = q^{-(2^h-1)}$$

by induction on h . For $h = 0$, this is $\Pr[S_0] = 1 = q^0$.

For $h \geq 1$, both subtrees have no fixed leaves and contain $r_1 = r_2 = 2^{h-1}$ random leaves. By the induction hypothesis and independence of the two subtrees,

$$\Pr[S_{h-1}^L \wedge S_{h-1}^R] = q^{-(r_1-1)} q^{-(r_2-1)}.$$

Conditioned on these two subtree survival events, the two output digits are independent and uniform on $\mathbb{Z}_q$. Hence the additional root survival condition has probability exactly $1/q$. Therefore

$$\Pr[S_h] = q^{-(r_1-1)} q^{-(r_2-1)} q^{-1} = q^{-(r_1+r_2-1)} = q^{-(2^h-1)}.$$

Finally, when no leaves are fixed, $r = 2^h \geq 1$, and the conditional uniformity of Ψ_h gives

$$\Pr[S_h \wedge \{\Psi_h = 0\}] = \Pr[S_h] \cdot \frac{1}{q} = q^{-(r-1)} q^{-1} = q^{-r}.$$

□

3.3 Second Moment for Injective Zero Candidates

Let Z_{inj} be the number of zero-admissible leaf-index assignments

$$\mathbf{a} = (a_1, \dots, a_k) \in [n]^k$$

whose indices are all distinct. Since Algorithm 2 examines all ordered candidate trees, the event $Z_{\text{inj}} > 0$ implies that Algorithm 2 finds a final candidate with value $0 \in \mathbb{Z}_q$.

Lemma 3.4 (First moment). *For $k = 2^L$,*

$$\mathbb{E}[Z_{\text{inj}}] = (n)_k q^{-k},$$

where $(n)_k = n(n-1) \cdots (n-k+1)$.

Proof. There are exactly $(n)_k$ ordered leaf-index assignments with all indices distinct. Fix one such assignment $\mathbf{a}$. Since the indices $a_1, \dots, a_k$ are distinct, the leaf values

$$X_{a_1}, \dots, X_{a_k}$$

are independent and uniform in $\mathbb{Z}_m$.

By definition, $\mathbf{a}$ is zero-admissible exactly when the height- L tree satisfies S_L and has final digit $\Psi_L = 0$. In this case no leaves are fixed and the number of random leaves is $r = k = 2^L$. Therefore the exact part of Lemma 3.3 gives

$$\Pr[\mathbf{a} \text{ is zero-admissible}] = \Pr[S_L \wedge \{\Psi_L = 0\}] = q^{-k}.$$

Summing this probability over the $(n)_k$ injective ordered assignments gives

$$\mathbb{E}[Z_{\text{inj}}] = (n)_k q^{-k}.$$

□

Lemma 3.5 (Second moment). *Let $n = \lfloor \alpha q \rfloor$, where $\alpha > 0$ is fixed. Then*

$$\mathbb{E}[Z_{\text{inj}}^2] \leq B_{L,\alpha},$$

where

$$B_{L,\alpha} := \sum_{s=0}^k \binom{k}{s}^2 s! \alpha^{2k-s}.$$

In particular, $B_{L,\alpha}$ depends only on L and α , not on q .

Proof. We expand

$$Z_{\text{inj}}^2 = \sum_{\mathbf{a}, \mathbf{b}} \mathbf{1}[\mathbf{a} \text{ is zero-admissible}] \mathbf{1}[\mathbf{b} \text{ is zero-admissible}],$$

where the sum ranges over ordered pairs of injective assignments $\mathbf{a}, \mathbf{b} \in [n]^k$.

Fix two such assignments $\mathbf{a}, \mathbf{b}$, and suppose they share exactly s initial sample indices. Let $A_{\mathbf{a}}$ and $A_{\mathbf{b}}$ denote the events that $\mathbf{a}$ and $\mathbf{b}$ are zero-admissible, respectively.

First expose the values with indices appearing in $\mathbf{a}$. Since $\mathbf{a}$ is injective, by the computation inside the proof of Lemma 3.4, it gives

$$\Pr[A_{\mathbf{a}}] = q^{-k}.$$

For every realization of these exposed values, the assignment $\mathbf{b}$ has exactly s fixed leaves, namely the leaves whose indices are shared with $\mathbf{a}$, and exactly $k - s$ remaining leaves whose values are still independent and uniform in $\mathbb{Z}_m$.

If $k - s \geq 1$, Lemma 3.3 gives

$$\Pr[A_{\mathbf{b}} \mid \text{the exposed values of } \mathbf{a}] \leq q^{-(k-s)}.$$

If $k - s = 0$, the same inequality holds trivially because its right-hand side is $q^0 = 1$. Therefore, pointwise over all exposed values,

$$\Pr[A_{\mathbf{b}} \mid \text{the exposed values of } \mathbf{a}] \leq q^{-(k-s)}.$$

Hence

$$\Pr[A_{\mathbf{a}} \cap A_{\mathbf{b}}] = \mathbb{E}[\mathbf{1}_{A_{\mathbf{a}}} \Pr[A_{\mathbf{b}} \mid \text{the exposed values of } \mathbf{a}]] \leq q^{-(k-s)} \Pr[A_{\mathbf{a}}] = q^{-(2k-s)}.$$

It remains to count pairs of injective assignments sharing s indices. There are at most $(n)_k$ choices for $\mathbf{a}$. Then choose the s positions of $\mathbf{a}$ that will be shared, the s positions of $\mathbf{b}$ receiving them, and the bijection between these positions; this gives at most

$$\binom{k}{s}^2 s!$$

choices. The remaining $k - s$ positions of $\mathbf{b}$ have at most n^{k-s} choices. Hence the number of such pairs is at most

$$(n)_k \binom{k}{s}^2 s! n^{k-s} \leq \binom{k}{s}^2 s! n^{2k-s}.$$

Thus

$$\mathbb{E}[Z_{\text{inj}}^2] \leq \sum_{s=0}^k \binom{k}{s}^2 s! n^{2k-s} q^{-(2k-s)} = \sum_{s=0}^k \binom{k}{s}^2 s! \left(\frac{n}{q}\right)^{2k-s}.$$

Since $n = \lfloor \alpha q \rfloor$, we have $n/q \leq \alpha$, and the stated bound follows. $\square$

3.4 Success Probability for the Actual Algorithm

Theorem 3.6 (Single-list success probability). *Fix $L \geq 1$, let $k = 2^L$, and fix $\alpha > 0$. Let q be an odd prime, set*

$$m = q^{L+1}, \quad R = q, \quad n = \lfloor \alpha q \rfloor,$$

and run Algorithm 2 on an initial list of n independent uniform samples from $\mathbb{Z}_m$. Then, for all sufficiently large q , Algorithm 2 returns a valid zero-sum k -tuple with probability at least

$$c_{L,\alpha} := \frac{(\alpha/2)^{2k}}{\sum_{s=0}^k \binom{k}{s}^2 s! \alpha^{2k-s}}.$$

In particular, the success probability is bounded below by a positive constant depending only on L and α .

Proof. By Lemma 3.4,

$$\mathbb{E}[Z_{\text{inj}}] = (n)_k q^{-k}.$$

Assume $q \geq 2k/\alpha$. Since $n = \lfloor \alpha q \rfloor$, for every $j \in \{0, \dots, k-1\}$ we have

$$n - j \geq \alpha q - 1 - j \geq \alpha q - k \geq \frac{\alpha q}{2}.$$

Therefore

$$(n)_k q^{-k} = q^{-k} \prod_{j=0}^{k-1} (n - j) \geq q^{-k} \left(\frac{\alpha q}{2} \right)^k = (\alpha/2)^k.$$

By Lemma 3.5,

$$\mathbb{E}[Z_{\text{inj}}^2] \leq B_{L,\alpha}, \quad B_{L,\alpha} = \sum_{s=0}^k \binom{k}{s}^2 s! \alpha^{2k-s}.$$

The Paley–Zygmund inequality, applied to the nonnegative integer-valued random variable Z_{inj} , gives

$$\Pr[Z_{\text{inj}} > 0] \geq \frac{\mathbb{E}[Z_{\text{inj}}]^2}{\mathbb{E}[Z_{\text{inj}}^2]} \geq \frac{(\alpha/2)^{2k}}{B_{L,\alpha}} = c_{L,\alpha}.$$

If $Z_{\text{inj}} > 0$, then there exists an injective zero-admissible leaf-index assignment as Algorithm 2 retains all survivors without subsampling, existence strictly guarantees discovery. By Lemma 3.2, this assignment appears in the final list with value $0 \in \mathbb{Z}_q$. For such a candidate, the reconstructed original sum is divisible by $q^{L+1} = m$, so the final verification step of Algorithm 2 accepts it. Therefore Algorithm 2 succeeds with probability at least $c_{L,\alpha}$. $\square$

References

- [1] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.